VISUD CHANG

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OBJECTIVE STATEMENT: Aerospace Engineering student with interests in mission operations, propulsion systems, and GNC. Seeking internship opportunities to apply technical and leadership experience to aerospace projects.

EDUCATION

University of California, Berkeley, August 2023–May 2027, B.S. in Aerospace Engineering, Minor in Electrical Engineering and Computer Science. GPA 3.94/4.00.

Relevant Coursework: Computer Science (Python, MATLAB, Java), Statistics and Data Analysis, Aerospace Design, Control Theory, Thermodynamics, Materials Science, Circuits, Orbital Mechanics, Propulsion, UAVs, Statics, Physics

Long Beach Polytechnic High School, PACE magnet program, graduated June 2023. Honors/awards: Math Department Pursuit of Excellence Award, AP Scholar with Distinction. Highest Honors, GPA 4.46/4.00

PROFESSIONAL EXPERIENCE

NASA Ames Research Center, *Moffett Field, CA, Jun 2025–Aug 2025, Mission Design Intern*

- Created MAESTRO, a Python/Dash geospatial tool that automates Mars rotorcraft landing zone selection using HiRISE terrain data and slope analysis.
- Integrated 3D visualization, slope mapping, and route planning to optimize airfield selection and reduce mission planning risk, expanding accessible terrain for science and exploration.
- Presented to NASA leadership how MAESTRO can enable more capable, autonomous planetary missions.

American Institute of Aeronautics and Astronautics, *Berkeley, CA, June 2024–present, Vice President*

- Oversaw chapter operations, event planning, and professional development initiatives for 25+ members
- Coordinated networking events, technical talks, and industry panels featuring aerospace professionals
- Led a team to execute resume workshops, interview prep sessions, and internship panels

Salas O'Brien, *Corona, CA, June 2024–August 2024, Mechanical Intern*

- Designed, developed, and reviewed the HVAC systems of over 20 buildings using AutoCAD and Revit
- Modeled 4 multi-level buildings and simulated airflow and cooling rates to improve rates while minimizing cost

PROJECT EXPERIENCE

Mission Simulator, *Personal Project, Feb 2025–Jul 2025*

- Built a spacecraft mission control and telemetry system with Python, Go, React, Arduino, C++, Git, and LoRa for real-time attitude/orbital tracking.
- Developed a TCP-based ground station and React GUI for live IMU telemetry, orbit visualization, and burns.
- Enhanced skills in full-stack development, networking, embedded systems, and simulation software design.

Regeneratively Cooled Rocket Engine, *Personal Project*, *Jul 2025–August 2025*

- Designed and simulated a regen-cooled liquid rocket engine in SolidWorks with symmetric coolant channels.
- Used SolidWorks Flow Simulation to optimize channel geometry in fluid, thermal, and pressurized systems, improving coolant distribution and reducing wall temperature by 30%.

Space Technologies and Rocketry (STAR), *Berkeley, CA, Sept 2023–present, Capacitive Fill Sensor Lead, Airframe and Propulsion Engineer*

- Optimized fin shape for UC Berkeley's first liquid engine rocket with Python/Ansys, increasing projected altitude by 15% while maintaining stability.
- Led design of STAR's first capacitive fill sensor using CAD/PCB, improving propellant measurement accuracy for a projected 5% apogee increase.
- Designed LOX tank mounts in SolidWorks and validated structural integrity through FEA, enabling the launch of Berkeley's first recoverable liquid engine rocket (May 2024).

SKILLS

Python, MATLAB, Java, Golang, Git, React, SolidWorks CAD, Arduino, C++, LoRa/RF communications, Docker, Computer Networking (TCP), OpenRocket, RocketPy, RasAero, AutoCAD, Pytest, Circuit Design, Microsoft Office